Design of an AI-driven Network Digital Twin for advanced 5G-6G network management

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***Abstract*—The Network Digital Twin (NDT) developed in the B5GEMINI project is presented in this article, highlighting its architecture, objectives, functionalities, and practical applications. The design of the NDT architecture is detailed, including the establishment of the foundational infrastructure, developed as part of B5GEMINI-INFRA. The integration of artificial intelligence techniques for network management tasks within B5GEMINI-AIUC is illustrated with relevant use cases, such as the detection of cybersecurity attacks and the simulation and optimization of virtual reality applications, to demonstrate the usefulness and potential of the proposed NDT solution. The platform enables controlled experimentation and data collection for training Machine Learning (ML) models, addressing challenges associated with realistic network traffic datasets and cybersecurity experiments without disrupting live networks. The infrastructure supporting the NDT allows for creating virtual scenarios, isolating traffic between experiments, on-demand traffic generation, and capture, ensuring repeatability and enabling evaluation of different detection and mitigation tools under identical conditions. Additionally, an in-depth use case focusing on ML-based detection of a simulated denial of service attack through DNS over HTTPS within a 5G network framework showcases the NDT's potential to provide a secure environment for testing and validating ML-based solutions without disrupting live networks.**

***Index Terms*—network digital twin, artificial intelligence, machine learning, cybersecurity, 5G network**

I. Introduction

A Digital Twin (DT) is a dynamic virtual model of a system, process, or service aimed at replicating the components, properties, and interactions of the entity with its environment [1]. Its primary purpose is to monitor and manage physical assets or systems in real-time or through scenario simulation to enhance industrial processes and foresee potential issues. Contrary to simulations, which focus on specific aspects and are based on pre-defined inputs, DTs encompass the entire system including physical components, receiving real-time data and enabling predictions of future performance. This concept extends to networks, where a Network Digital Twin (NDT) emulates a real network providing a virtual representation as defined by the Internet Engineering Task Force (IETF) [2]. Efforts are underway to establish a standardized framework for NDTs, facilitating real-time mapping between physical and virtual networks and aimed at analysis, diagnosis, emulation, and control of real networks.

Considering the ideas outlined by the IETF, in this article, we describe the design and development of an NDT architecture carried out in the context of the B5GEMINI project, which is divided into two subprojects: B5GEMINI-INFRA and B5GEMINI-AIUC. B5GEMINI-INFRA aims to establish the foundational infrastructure for NDTs, providing the necessary framework and resources for their deployment and operation by incorporating key components such as data collection mechanisms, virtualized and non-virtualized network emulation, and real-time network interaction interfaces.

On the other hand, B5GEMINI-AIUC integrates Artificial Intelligence (AI) techniques, including ML and Deep Learning (DL), into the NDT framework for various network management and optimization tasks. This includes designing and implementing tailored ML/DL models for tasks such as resource allocation, traffic optimization, and anomaly detection, along with ensuring their effectiveness and reliability through training and validation procedures within the NDT environment. To demonstrate the practicality of the NDT architecture, B5GEMINI-AIUC conducts diverse use cases and experiments, showcasing its versatility in different network environments through scenarios ranging from cyberattack detection to virtual reality application optimization.

The infrastructure proposed by B5GEMINI-INFRA is capable of hosting the AI-based NDT, aiming to deploy a 5G/6G NDT support platform for configuring custom network setups and generating traffic for data collection. This collected data supports ML and AI-driven analysis for cybersecurity and network management scenarios. Goals include resource optimization and cybersecurity measures within beyond 5G (B5G) communication networks. The NDT architecture extends to other use cases as well. It serves as an experimentation platform for testing deep reinforcement learning technology for multimedia content cache selection, optimizing resource deployment without affecting the real network. Additionally, it serves as a development platform and decision-making tool for extended reality architectures that go beyond traditional reality experiences, encompassing virtual reality, augmented reality, and mixed reality, offering interactive and enhanced experiences blending physical and digital worlds.

This work proposes a technological ecosystem centered around a core network using 5G/6G technology for repeatable NDT experiments. It emphasizes two pillars: an open-source

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platform for implementing the NDT ecosystem and collecting realistic data and an AI-based application layer, which utilizes this data to address operator and manufacturer needs in network management, resource optimization, and cybersecurity in 5G/6G networks. Existing literature on NDT applications for advancing 5G, beyond 5G, and 6G networks mainly focuses on prototyping networks to emulate an environment during the design phase, facilitating informed decision-making and the subsequent transfer of this acquired knowledge to operational networks [3], [4]. Our contribution aims to extend this by addressing challenges in generating network traffic datasets in operational networks, such as restrictions from operators and privacy concerns. Emulating cybersecurity attacks on real networks poses risks such as service disruptions, which can adversely impact end users. In B5GEMINI-AIUC, we propose generating synthetic traffic data for AI model development, using NDT as a controlled experimental environment for this purpose. This allows for the training and development of AI models using synthetic traffic data from cybersecurity attacks on next-generation networks.

To exemplify the potential of the proposed AI-driven NDT in addressing cybersecurity challenges within 5G networks, in this article, we present a use case focusing on the detection and mitigation of a DNS over HTTPS denial of service attack, which involves encrypted traffic and cannot be effectively detected through conventional methods such as packet inspection [5]. In this use case, we show how the integration of AI in the NDT enables preemptively identifying and responding to malicious activities targeting network infrastructure such as DNS servers. In this context, the NDT provides an intermediary platform for validating decisions generated by the ML model before their implementation in the live network environment, ensuring uninterrupted network availability for legitimate users. Additionally, this use case highlights the NDT as a controlled experimental environment for developing and validating AI-based cybersecurity solutions, facilitating the training and continuous refinement of ML models under realistic conditions without disrupting operational networks.

This article is organized as follows. Section II presents a review of related work. Section III outlines the key components, objectives, and functionalities of the proposed NDT solution. Section IV describes the architecture and infrastructure of the proposed NDT architecture. Section V focuses on the practical application of the NDT, specifically on the Encrypted Traffic Use Case developed in the context of the B5GEMINI project. Finally, Section VI concludes the article by summarizing the main conclusions of our work and discussing interesting future directions.

II. Related Work

A review of the literature concerning NDT solutions and their applications for network management and optimization tasks through the use of AI is presented in this section.

A. Network Digital Twins

As explained in the previous section, the DT paradigm can be applied to a vast number of areas, including communication networks. These types of networks are ideal candidates for the application of DT technology to improve existing solutions due to the increase in the number of companies that use them, their scale, or their complexity. Since early 2017, research has been conducted on the application of this paradigm to communication networks; some relevant work in this field is cited below. In academia, the authors in [6] apply the DT paradigm in 5G networks with multi-access edge computing. In this context, the DT is used offline with the aim of implanting algorithms through reinforcement training to optimize the resource allocation and energy savings of the associated hardware components. In [7], the authors set up an edge network in the form of a DT, in which they deploy a server in the form of a DT that evaluates the state of the physical server. In this way, the edge network of the DT provides data for the training of offloading strategies.

Another use case is the one presented by the authors in [8] in 2021, which involves the use of DTs in complex 5G networks. Due to the special characteristics of this type of application to communication networks, it is necessary to develop platforms that differ greatly from some frameworks or platforms presented in the previous section. Therefore, in [8], the authors present a platform geared towards automated management and intelligent integration of data center networks, proposing the simplification of workflows associated with network services management. Efforts to apply DT technology in communication networks are still in early research stages. Presently, it's mainly applied in specific scenarios such as network optimization, acting as a simulation tool for problem-solving in network operation and maintenance. However, a comprehensive approach to integrating DTs with real networks across their lifecycle is essential to enable future applications in planning, deployment, maintenance, and optimization tasks, enhancing automation and network intelligence levels.

The NDT proposed in this work is based on two pillars: The Network Functions Virtualization/Software-Defined Networking (NFV/SDN) [9] paradigm and the use of open-source technology. The inclusion of virtualized components in network topologies is becoming more prevalent, especially within the realm of 5G and beyond standards, along with advancements in SDN and NFV technologies. These technologies provide several benefits, such as agility, flexibility, cost-effectiveness, scalability, interoperability, and vendor independence. Open-source technologies offer cost-effectiveness by providing free software access and reducing licensing expenses. The code transparency enhances security and encourages a collaborative community for ongoing support and innovation. Moreover, the flexibility, interoperability, and avoidance of vendor lock-in make open-source solutions attractive for organizations seeking efficient and adaptable technology. Thus, below is a compilation of open-source tools facilitating the incorporation of both virtualized and non-virtualized functionalities for developing the NDT.

- Kubernetes [10]: It is an open-source container orchestration platform designed to automate the deployment, scaling, and management of containerized applications.

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Kubernetes provides a robust and scalable framework for container orchestration, allowing users to efficiently manage and coordinate containerized workloads across a cluster of machines.

- Kubernetes Network Emulator [11]: Various network emulation approaches utilize virtual machines to simulate hardware routers. These systems are effective in validating vendor control plane implementations and limited data plane certifications. This project aims to establish a standard "interface" for vendors to create container implementations, facilitating the construction of intricate network topologies.
- Containerlab [12]: Provides a command line interface for orchestrating and managing container-based networking labs. It initiates containers, builds virtual wiring between them to create lab topologies of the user's choice, and manages the lifecycle of the labs.
- Vrnetlab [13]: Packages a regular virtual machine inside a container and makes it runnable as if it were a container image.
- MetaLB [14]: Serves as an implementation of a load balancer for network environments.
- free5gc [15]: It is an open-source project for 5th generation (5G) mobile core networks. The ultimate goal of this project is to implement the 5G core network defined in 3GPP Release 15 and beyond.
- STGUTG (Signaling Traffic Generation/User Traffic Generation) [16]: It is a software created for the generation of both signal and user traffic to be sent to a 5G network core.
- Open Source MANO (Open Source Management and Orchestration, OSM) [17]: It is an open-source project that focuses on developing a Management and Orchestration (MANO) stack for NFV, which involves the virtualization of network functions traditionally performed by dedicated hardware appliances.
- OpenStack [18]: It is an open-source cloud computing platform that provides a set of services for building and managing both public and private clouds. It aims to offer infrastructure as a service by enabling users to control and provision compute, storage, and networking resources through a centralized dashboard.
- Hardware equipment such as traffic generators.

B. AI Applications within Network Digital Twins for 5G Network Management

In [19], AI is described as a strong candidate to help automate and improve the decision-making of NDTs in the context of their deployment in an Industry 4.0 setting. This same work describes the multiple roles that AI could fulfill, providing services such as dynamic resource scaling, intrusion detection, and Radio Access Technology selection to fulfill specific SLAs and KPIs. Thus, AI agents and ML algorithms enable optimizations in reliability, robustness, and performance in DTs. The integration of AI techniques can have multiple uses in the context of DTs. As mentioned above, DL models can be trained to optimize resource allocation and thus minimize energy consumption as in [6].

Another field with great potential for AI NDT applications is cybersecurity. In [20] and [21], the authors create an NDT oriented as a cyber range platform to train cybersecurity experts powered by ML tools. This tool allows for the creation of complex scenarios that emulate attacks using synthetic traffic and educates learners by making them analyze the performance of pre-trained ML models created for the detection of attacks within the NDT. Other works such as [22] use DL algorithms for intrusion detection, which are combined with Software Defined Networking and DT technology to allow safe communication within an insecure open channel scenario. Intrusion detection assisted by AI has also been shown to be effective within a DT in industrial control systems [23]. In [24], the authors determined that the use of a DT approach using sensors and ML techniques in additive manufacturing enables the detection of potential anomalies caused by malicious cyber-physical intrusions.

The full integration of AI components for cybersecurity is contemplated in [25], [26] and [27], where the authors propose the creation of an AI-powered NDT for 5G and beyond networks. A principal cornerstone of this project is the Mouseworld Lab [28], which is a controlled environment that allows the creation of complex network scenarios. Within this environment, ML models can be freely trained and validated on realistic data for multiple use cases, such as cybersecurity, network management, and dynamic deployment of NDT instances.

III. Design and Development of an AI-powered Network Digital Twin

The research and development executed under the B5GEMINI project and described in this work aims to advance the adoption of innovative AI and ML techniques in an NDT environment for advanced 5G and 6G networks management tasks. The NDT platform supports virtualization and softwarization mechanisms, enabling emulation of advanced operating environments based on 5G and beyond. Additionally, the platform accommodates the deployment of virtual instances known as slices over a common infrastructure. Security is prioritized, with the platform providing a secure environment for deploying cybersecurity experiments without impacting the physical twin.

The platform supports experiments involving both Internet or external network access and internal communication within the B5G network. It is capable of reproducing specific network situations, enabling performance measurements in response to different policies or decision-making considerations based on factors such as monitored traffic, network congestion, or new functionalities. The platform also includes data collection mechanisms to evaluate network performance during experiments and capture datasets essential for training ML algorithms. The architecture's versatility is enhanced by the platform's capability to generate traffic that adapts to dynamic load patterns and network conditions. Moreover, it supports

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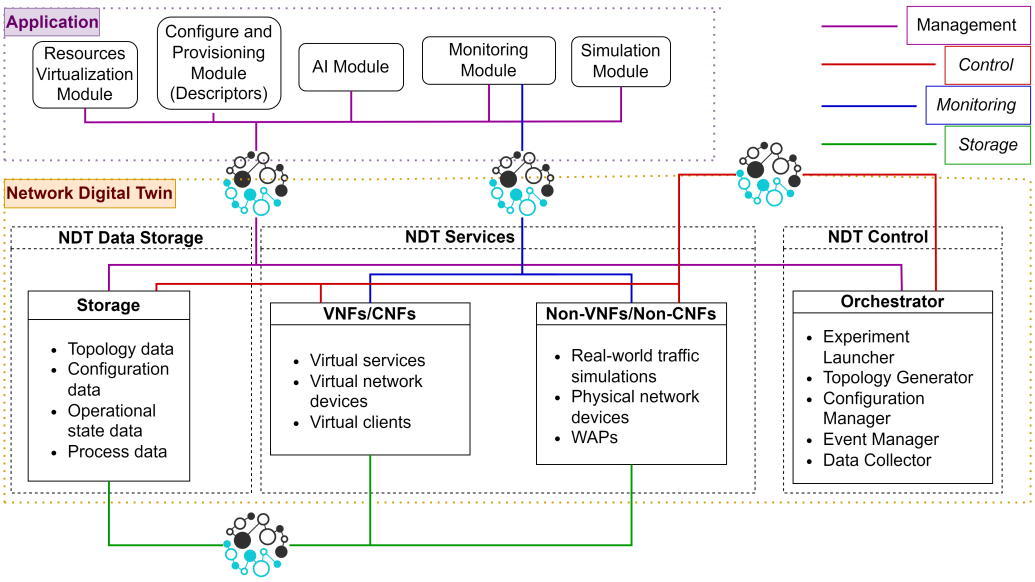


Fig. 1: Schematic representation of the main layers of the network digital twin infrastructure.

traffic injectors in B5G environments, aiding the integration of existing applications on the B5G infrastructure.

Through the data collected in the NDT deployments carried out on the platform, the AI-based applications aim to achieve the objectives described below. The first objective involves the investigation and development of generative models based on Generative Adversarial Networks for obtaining high-quality user and network traffic data that can entirely replace real data in ML tasks of the NDT. In parallel, efforts are directed towards researching, designing, and implementing techniques that enhance the resilience of ML/DL models against recently emerged adversarial attacks.

The research agenda includes the exploration and development of algorithms and mechanisms inspired by Green-AI and AutoML techniques. The primary objective is to devise methods for selecting the most effective features, ensuring the efficiency of compact ML and DL models deployed in a NDT. The platform aims to design, deploy, and validate a series of use cases. Firstly, a cybersecurity use case where encrypted traffic related to attacks will be identified using ML/DL techniques. The next use case presents a social virtual reality scenario where the influence of latency on the interaction between two or more people in a cooperative virtual environment will be evaluated. Lastly, attention is turned towards multimedia transmision, where in the last use case network parameters will be optimized using the ALTO protocol through ML and reinforcement techniques.

IV. Network Digital Twin Infrastructure

The infrastructure presented in this work, based on the NFV/SDN architecture, has the capability to create instances of virtual scenarios, isolate traffic between scenarios in experiments, generate on-demand network traffic, and capture it. A key feature is its repeatability, where identical environmental conditions allow the evaluation of different mitigation tools or versions based on similar statistical patterns. As illustrated in Figure 1, the infrastructure consists of two main layers.

- **Application Layer:** This layer encompasses tools and technologies enabling developers to create and deploy applications and services running on the virtualized network infrastructure, interacting with scenarios in the DT.
- **NDT Layer:** Comprising functional components of the DT of the network.
  - **NDT Data Storage:** Housing data and/or real-time data (configuration data, operational state data, topology data, tracing data, metric data, process data, etc.) about its real-world twin, as outlined in the framework proposed by the IETF. Data are characterized as the sole source of truth and stored in the data repository, supporting the construction of various models.
  - **NDT Services:** Comprising network elements intended for emulation, including both virtualized and non-virtualized components such as hardware equipment. Notable features of the NDT Services in the infrastructure include:
    - NFV orchestration based on OpenStack, OSM, and Kubernetes.
    - Automatic deployments based on templates.
    - Multiple slices or virtual experiments.
    - Integration with traffic generators.
    - Availability of tools for PCAP injection onto the NDT.
    - Internet connectivity (service and management): for non-replicable services locally with network switch interconnectivity.
    - Emulation of 5G network for transport.
  - **NDT Control:** The control module performs critical actions to effectively and efficiently manage the

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network, including deployment, configuration, and administration of different functional modules.

Communication between both layers, as well as among their contained components, occurs through the implemented Data Fabric connecting various twin components. This allows the integration and organization of data from disparate sources into a knowledge graph with a common format. These data can be distributed to any system, establishing bidirectional connections between data sources capable of assuming both producer and consumer roles.

V. Encrypted Traffic Use Case and Experiments

This section presents the design and deployment for an encrypted traffic scenario using the NDT. After an introduction of the use case, various aspects including its design, system architecture, deployment and testing procedures, training and validation of ML/DL, and its integration within the B5GEMINI-INFRA framework will be detailed.

This use case involves simulating a denial of service attack through DoH (DNS over HTTPS) flooding within a 5G network framework using the NDT. DoH is a protocol that encrypts DNS queries and transmits them through HTTPS channels, enhancing privacy and security [29] by preventing third parties from intercepting DNS traffic and potentially manipulating DNS responses. The attack aims to overload a DoH server with excessive requests by using a high-volume application in the upstream direction, disrupting DNS resolution and causing denial of service for legitimate users [30]. Emulating this attack in the NDT framework requires the deployment of a 5G network architecture.

Conventional rule-based Network Intrusion Detection Systems (NIDS), such as Snort, are ineffective in detecting and mitigating attacks involving encrypted protocols such as DoH. Even for unencrypted DNS flooding attacks, traditional defensive measures are often inadequate and easily circumvented [5]. To address this challenge, ML-based methods are employed in this use case to detect and respond to DoH flooding attacks. The data to train these ML models will be produced by emulating the DoH attack scenario and collecting various types of data. The variables to be measured include traffic data at different significant network points and resource utilization data (such as CPU, memory, and I/O) on the deployed network functions.

Utilizing NDT for this specific use case offers significant advantages for enhancing cybersecurity measures within 5G networks and enabling secure experimentation. With a DT, organizations can conduct safe and cost-effective experiments in a virtual environment, testing new configurations, protocols, or services without disrupting the actual network. This allows engineers and researchers to explore innovative ideas and assess their impact before implementation. In the Encrypted Traffic Use Case, having a platform for conducting cybersecurity experiments is highly advantageous for generating realistic data without disrupting live network operations while completely avoiding data privacy issues. This facilitates the development of datasets used in training and validating ML models for decision-making within the real network.

A. Use Case Design

Figure 2 illustrates the detailed topology of the scenario. In this 5G architecture, both the control plane, the data plane, and the radio access (gNodeB, gNB) are represented by virtualized network functions connected to the transport network. The gNB is connected to multiple end-user clients, while the User Plane Function (UPF) has Internet connectivity. During the experiment, we will monitor the links in the transport network, the routers, and the resource utilization of the control plane and data plane modules. To carry out the attack, we will inject heavy-hitter traffic using a synthetic traffic generator running on virtualized clients to the DoH server to disrupt the system.

B. Deployment and Testing of the Use Case

The NDT scenario simulates a heavy-hitting application in the upstream direction, such as remote cameras or streaming services, triggering a flooding attack against a DoH server on a mobile operator's network. The attack employs the DoH flooding protocol, generating an overwhelming volume of encrypted requests, disrupting the DNS server and potentially affecting other network components such as the UPF. This emulation aims to extract variables for training ML models to filter and detect malicious traffic [31]. These variables include traffic data at various network points and resource utilization data (e.g., CPU, memory, and I/O) on deployed network functions.

To deploy this topology, a descriptor in YAML format was previously extracted. This descriptor reflects the topology represented in Figure 2 and intuitively displays the different modules of the topology along with their corresponding container images, connections, and configuration file paths for each. Since this is an exploratory phase, both the metrics to be extracted from the experiment and the architecture of the scenario itself are subject to change in future iterations. The selected case is a first approximation, and it will evolve depending on factors such as technical feasibility and security evolution according to network requirements or needs identified during the evolution of the 6G vision.

C. Training and Validation of the ML/DL Models

To effectively protect against DNS flooding attacks, it is crucial to prevent harmful traffic from reaching specific DNS servers. As stated previously, conventional NIDS techniques such as domain or IP blocking are ineffective against DoH due to the encrypted nature of the traffic. To address this issue, building upon the methodology detailed in our previous study [5], a real-time detection system is proposed for DoH flooding attacks at the network level using ML models based on a comprehensive set of network flow statistics used as input features for the models. Currently, we are awaiting the dataset generated in the NDT needed for training and validation of the model, which will be initiated once the data collection phase of the use case deployed in the NDT is completed.

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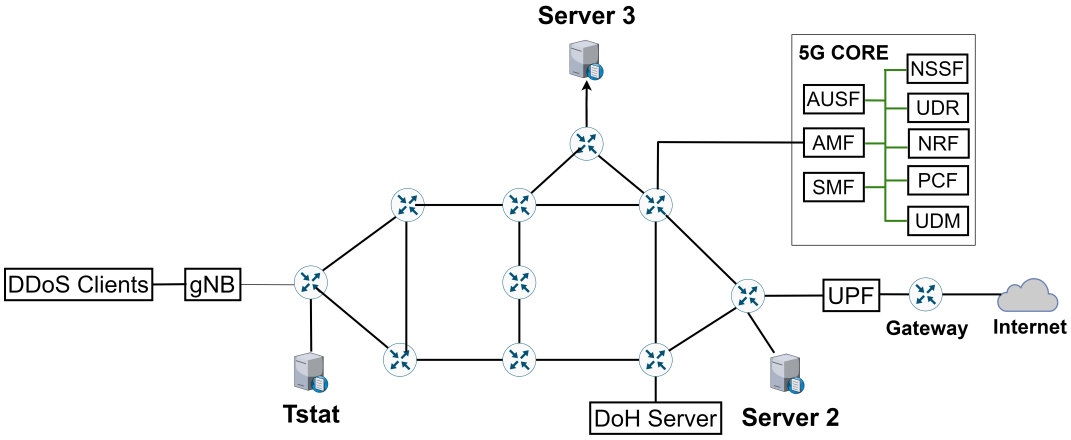


Fig. 2: Design of the topology employed for the deployment of the Encrypted Traffic Use Case.

The dataset will be processed using the Tstat tool to extract statistics from aggregated network flows. Three classes are defined for network flows: 0 (Normal traffic), 1 (Standard DoH traffic), and 2 (DoH flooding attack traffic).

ML models will undergo supervised learning using labeled network flow data to iteratively adjust their parameters and optimize classification performance. The dataset will be partitioned into training, validation, and testing subsets to prevent overfitting. Various algorithms such as decision trees, and neural networks will be evaluated to determine the most suitable model for accurately detecting DoH flooding attacks. Once trained, models will be deployed in the NDT to monitor real-time network traffic. Upon detecting anomalous behavior consistent with a DoH flooding attack, appropriate mitigation measures will be initiated promptly. These may include rate limiting or traffic redirection, chosen based on the model's confidence level and anticipated impact on the network.

To ensure safety and reliability, mitigations will first be applied within the NDT and then transferred to the live network infrastructure to prevent DoH flooding attacks from affecting DNS servers following validation, ensuring thorough testing before implementation. This approach minimizes the risk of disruptions to legitimate network operations. Furthermore, the integration of the NDT allows for feedback mechanisms to refine detection capabilities and decisions based on real network outcomes and observed changes in attack behaviors. Additionally, continuous monitoring and periodic retraining of ML models ensure adaptability to evolving attack strategies and network conditions, enhancing DNS infrastructure resilience against emerging threats. Integration with the NDT facilitates this adaptation by providing an accurate representation of the real network, simulating diverse traffic patterns, and dynamically adjusting attack complexity. Operating within a controlled environment enables accurate data labeling, crucial for constructing datasets for model training and validation.

VI. Conclusions and Future Work

In conclusion, the development and implementation of a Network Digital Twin (NDT) architecture offers a versatile platform for emulating, analyzing, and optimizing network environments through the use of Artificial Intelligence (AI). B5GEMINI-INFRA project establishes foundational infrastructure for NDTs, enabling the deployment and operation of virtual network replicas with real-time interaction capabilities in a secure environment for reproducible experiments, without impacting the real network. B5GEMINI-AIUC project integrates AI techniques to enhance cybersecurity solutions within the NDT. Diverse use cases demonstrate the NDT's utility across various realistic scenarios, showcasing its versatility. Notably, the NDT accurately emulates real-world network conditions, facilitating the creation of realistic synthetic traffic datasets for ML model training. In the Encrypted Traffic Use Case, NDT-trained ML models classify and mitigate cyber threats in real-time, ensuring 5G network resilience. The NDT serves as an intermediary platform for validating ML model decisions before live network implementation, ensuring uninterrupted service for legitimate users.

Future work aims to expand the detection capability of the ML components to a broader spectrum of cyber threats beyond DDoS attacks. Exploring adaptability to network variations via reinforcement learning holds promise for dynamically adjusting NDT to evolving traffic patterns. Additionally, research should address the scalability of system components for maximum efficiency across networks of varying sizes. Further ideation of use cases could enhance the versatility of the NDT architecture. Finally, the scalability and performance of the NDT architecture will be evaluated to support large-scale multi-domain network deployments and heterogeneous network environments.

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